



Metagenomics insight into Puga geothermal geyser located in Himalayan Geothermal Belt (Trans-Himalayan Plateau) Ladakh, India

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Abstract

Puga geothermal geyser and surrounding area, located in the Himalayan Geothermal Belt of the Trans-Himalayan Plateau in Ladakh, India, are very geographically isolated and considered pristine and free of anthropogenic activities. In this study, we have conducted the first metagenomic investigation of the microbes in and around the geyser. The whole genome sequencing analysis showed the presence of a total of 44.8%, 39.7% and 41.4% bacterial phyla in the PugW, PugS, and PugSo samples respectively, 8.6% of archaeal phyla (in all the samples), unclassified (derived from other sequences, PugW: 27.6%, PugS: 27.6%, and PugSo: 15.5%) and unclassified (derived from bacteria, PugW: 12%, PugS: 13.8%, and PugSo: 13.8%). The majority of archaeal sequences were linked to Euryarchaeota (2.84%) while the majority of the bacterial communities that predominated in most geothermal locations were linked to Pseudomonadota (67.14%) and Bacteroidota (12.52%). The abundant bacterial strains at the species level included *Dechloromonas aromatica*, *Acinetobacter baumannii*, and *Arcobacter butzleri*, in all the samples while the most abundant archaeal species were *Methanosaeta thermophila*, *Methanoregula boonei*, and *Methanosarcina berkeri*. Further, this geothermal geyser metagenome has a large number of unique sequences linked to unidentified and unclassified lineages, suggesting a potential source for novel species of microbes and their products. The present study which only examined one of the many geothermal geysers and springs in the Puga geothermal area, should be regarded as a preliminary investigation of the microbiota that live in the geothermal springs on these remote areas. These findings suggest that further investigations should be undertaken to characterize the ecosystems of the Puga geothermal area, which serve as a repository for unidentified microbial lineages.

Keywords Himalayan Geothermal Belt · Natural geothermal geyser · Extremophiles · Microorganisms · Metagenomics studies · Functional analysis

Introduction

In the past two decades, a myriad of extreme environments has been extensively studied for their microbial diversity, including hot springs [1–2], hydrothermal vents [3], fumaroles/stream vents [4], and volcanic soils [5]. Cultivation-independent methods are essential for understanding

microbial diversity since most microorganisms in such extreme conditions are non-cultivable [6]. Investigations into the microbiological or geomicrobial richness of hot springs have resulted not only in major scientific breakthroughs [7] but have also enriched our understanding of life in extreme conditions [8]. However the primary emphasis of microbial diversity investigation has been on low-elevation hot springs such as Kamchatka, Russia [9], Iceland [9], Yellowstone National Park [10], Long Valley Caldera [11], and various regions in China [1]. Except for Tibetan geothermal hot springs from the Himalayan Geothermal Belt, there is not a significant amount of data on the diversity of microbes of high-elevation hot springs worldwide [7]. The geothermal springs and geysers serve as intriguing analogs for the early history of the biogeosphere because they sustain microbial populations at high temperatures in an ecosystem with a high potential for preservation [12]. Given that,

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unlike terrestrial ecosystems, geothermally heated springs and geysers offer distinct aquatic environments with significant geochemical and physical gradients scattered over close and distant geographic areas, they are good research platforms for microbial biogeography. Because geothermal springs have fewer microbial community structures than soils and sediments from other areas, robust identification of diversity patterns is possible.

The Himalayan Geothermal Belt is one of the biggest geothermal locations on Earth, with diversified contours, canalculated geology, surreal valleys, and pulchritudinous sceneries. This belt encompasses the entire Himalayas and extends over 3,500 kilometers from Pakistan, Tajikistan, Afghanistan, the North West Indian Himalayas (regions of Uttarakhand, Ladakh, Jammu-Kashmir and Himachal Pradesh), Bhutan, Nepal, the North East Indian Himalayas (regions of Sikkim, Assam, Meghalaya, and Arunachal Pradesh), Tibet, Thailand, Myanmar, and the Yunnan Provinces of China [13]. This belt is reminiscent of the enormous geotectonic collision that occurred when the Indian Plate and the Eurasian Plate merged [14]. Several geothermal zones have been identified in the Indian Sub-Continent. However, the Puga geothermal field (33°13' N and 78°18' E), which is situated in the northwest part of the Himalayas [15], is thought to be the most promising. It is located in the Union Territory of Ladakh (32°N to 36°N and 75°E to 80°E) in India and makes up part of the Himalayan geothermal belt [16]. The Puga region is surrounded by hills that ascend to a height of approximately 6000 m, forming a valley there

between Polokongka La in the west and Sumdo settlement in the east, a valley of 15 km length and 1 km width follows in an east-west direction (Fig. 1). This region, which is south of the Indus Suture Zone, has approximately hundred hot springs and geysers, with surface flow temperature reaching 84°C (the boiling point of water at that altitude). Borax and sulphur, which are genetically related to thermal fluids, are widely distributed in the eastern part of the valley [17]. The water, sediment, and soils from the Puga natural geothermal geyser (33°13'23" N and 78°18'54" E [18], differ from those of the other hot springs and geysers because they contain higher concentrations of boron and sulfur.

Furthermore, the high mineral concentration, low microbial load, physically and chemically diverse sample of the high-elevation geothermal geyser make it difficult to recover unbiased metagenomic DNA for a study of the diversity of the microorganisms present there. When soil metagenomic DNA is extracted, several metals, including boron and sulfur, can co-extract as contaminants and significantly impact the efficacy of cell lysis [5]. Because Puga's geothermal geyser has a high temperature in the middle of a cold desert and hasn't been investigated using metagenomics up until now, and to the best of our knowledge, this is the first report ever, we hypothesized that the diverse population of microbial extremophiles present in the geyser could be used for biotechnological and industrial purposes. We used Illumina technology for high-throughput sequencing in the present investigation to carry out whole-genome functional analysis on metagenomic sequences as well as deep sequencing of

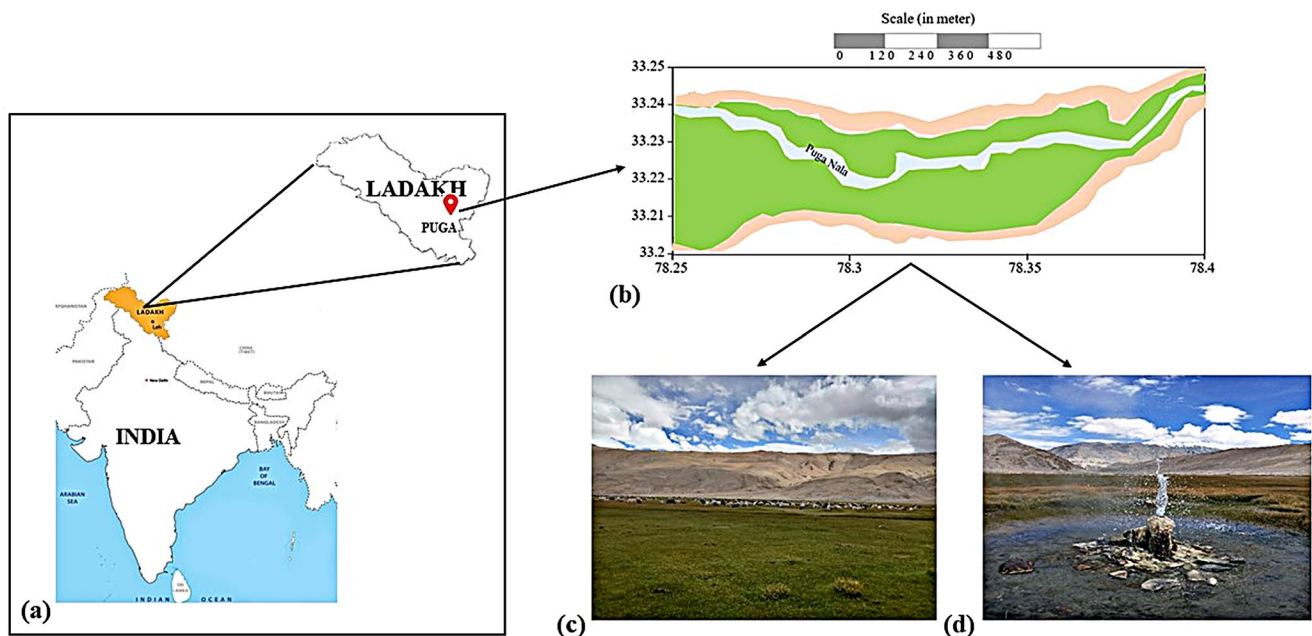


Fig. 1 Location sites (a) schematic map showing the location of the study site in the Western Indian Himalayan Geothermal Belt (Ladakh, 32°N to 36°N and 75 °E to 80°E) region, (b) map of Puga Geothermal

Field (33°13' N and 78°18' E), Ladakh, India, (c) Puga Geothermal Area, and (d) Puga Geothermal Geyser (33°13'23" N and 78°18'54" E)

the metagenomic microbiome from Puga natural geothermal geyser water, sediment, and soil samples. The samples were collected from the Puga geothermal geyser and near about areas. The finding will improve our knowledge of the site's genomic characteristics and decipher the microbiological diversity, metabolic profile, and functional characteristics of the geothermal aquatic bioresource, producing a metagenomic wealth for identifying and characterizing novel genes encoding biocatalysts with industrial potential.

Materials and methods

Site description and sample collection

Four water (PugW), three sediment (PugS), and three soil (PugSo) samples were collected from the Puga natural geothermal geyser and its nearby area in the Himalayan Geothermal Belt (Trans-Himalayan Plateau), Ladakh, India on September 10, 2021. Sediments are layers or collections of particles that have been removed from the location where they were initially weathered from rock and redeposited elsewhere. In contrast, soils are results of the earth's crust weathering in situ. All samples were collected in a sterile thermostable sample container, stored in iceboxes, and later transferred to 4 °C until further analysis. Supplementary Table 1 shows the physicochemical parameters of the Puga geothermal geyser. The geyser temperature ranged from 45°C to 84°C, with pH levels ranging from 8.3 to 8.6. Total dissolved solids (measured in ppm) ranged from 129 to 822. All of these measurements were taken on-site in real time.

Isolation and quantification of environmental DNA

The samples of water (04), sediment (03), and soil (03), respectively, were pooled, and the pooled samples of water (01), sediment (01), and soil (01) were used for the metagenomic study. Environmental DNA was recovered from water samples using the DNeasy PowerWater Kit (Qiagen, Hilden, Germany), while sediment and soil samples were isolated using the DNeasy® PowerSoil® kit (Qiagen, Hilden, Germany). The amount of isolated DNA was quantified using a Nanodrop (ThermoFisher Scientific, Massachusetts, USA) and gel electrophoresis. Final libraries were measured using a Qubit 4.0 fluorometer (ThermoFisher Scientific, Massachusetts, USA, #Q33238) and a DNA HS assay kit (ThermoFisher Scientific, Massachusetts, USA, #Q32851) following the instructions provided by the manufacturer.

Whole-genome shotgun metagenome sequencing and data analysis

The library was sequenced using the Illumina NovaSeq 6000 platform (Illumina, USA). The insert size of the library was determined using the TapeStation 4150 (Agilent Technologies, Palo Alto, CA, USA) and high sensitive D1000 screentapes (Agilent # 5067–5582) in accordance with the manufacturer's methodology. FastQC v.0.11.9 was used for assessing the quality of the raw fastq reads from the sample [19]. The raw fastq reads have been processed using Fastp v.0.20.1 [20], then quality checked with FastQC and summarised with MultiQC [21]. The processed reads were submitted to taxonomic assignment using MG-RAST [22] against the Silva SSU Databases at six taxonomic lineage levels: phylum, class, order, family, genus, and species. The processed readings were then subjected to Taxonomic Classification using KRAKEN2 [23]. The OTU file in BIOM format was generated from the KRAKEN2 output using the tool Kraken-boom (<https://github.com/smdabdoub/kraken-biom>). KronaTools (<https://github.com/marbl/Krona>) was used to generate Krona Plots of the individual samples. The merged OTU table in biom format was submitted to AMPVIS2 to generate the OTU Clustering Heatmaps, PCA plot, and Alpha Diversity plots [24]. Post Analysis Queries: Based on the top 50 average abundance of the OTUs, phylogenetics was constructed using the NCBI common tree [25] and visualised using Interactive Tree Of Life (itol) [26]. The data are publicly available in NCBI Sequence Read Archives (SRA) Accession Nos: SRR24760245 (soil), SRR24760246 (sediment), and SRR24760247 (water), with BioProject ID. No.: PRJNA977122 and BioSample ID. No.: SAMN35453230.

Results and discussion

Physicochemical parameters of the sampling site

The borehole, geyser, and spring discharge chemicals from the Puga geothermal system are remarkably homogeneous. The unequivocal indication that a single source is responsible for thermal discharges was provided by the consistent abundances of non-reactive ions like chlorine, boron, and fluorine [27]. The Puga fluid is an abundant blend of common and rare alkalis with bicarbonate-chlorine ($\text{HCO}_3\text{-Cl}$) type water and is moderately diluted (TDS~2400 mg/l). The silicon dioxide, boron, and fluorine concentrations are relatively high, and the molal sodium/potash (Na/K) is around 13. It is unique since it is the only known geothermal fluid with a caesium > lithium ratio and has the highest relative caesium concentration [27–28]. Carbon monoxide,

hydrogen sulfide, ammonia, nitrogen, methane, and helium are the most abundant gases in thermal discharge, in the same order [29]. The average concentration of elements like barium, strontium, lead, zinc, and copper, in the Puga fluid were evaluated to be approximately equal to 0.2, 0.5, 1×10^{-3} , 4×10^{-2} , and 5×10^{-3} mg/l, respectively [30].

Microbial taxonomic diversity

A total of 10 samples i.e. water (04), sediment (03), and soil (03) were collected from the Puga geothermal geyser, Ladakh, and pooled samples of water (01), sediment (01), and soil (01) were used for DNA sequencing analysis to recognize novel microorganisms. These samples were all obtained in the vicinity of the Puga geothermal geyser in Ladakh, India. These samples were prepared by creating DNA libraries for Illumina sequencing, and the sequence data obtained from this technique was examined using informatics pipelines (MG-RAST and KRAKEN2) designed to detect known or novel microorganisms (Table 1). KRAKEN2 was used for taxonomic classification and MG-RAST was used for taxonomic, functional analysis, alpha diversity and, phylogenetic analysis. A total of 101,275,022, 81,719,124, and 92,235,980 reads, respectively were obtained from the MG-RAST and KRAKEN2 system as raw data, with an average length of 159 bp in all the samples i.e. water (PugW), sediment (PugS) and soil (PugSo). The metagenomic data were put together using Newbler, a program that makes algorithmic attempts to join shorter contigs together from individual sequence reads. For bioinformatics analysis, a total of 88,749,234 (PugW), 70,790,984 (PugS), and 90,785,094 (PugSo) readings that passed MG-RAST quality filters were

employed (Table 1). The trimmed read length (in bp) were 136 (PugW), 127 (PugS), and 149 (PugSo) and the average GC (guanine-cytosine) content across all contigs were 55% (PugW), 49% (PugS), and 55% (PugSo).

The average GC content in each assembly of eight widely dispersed terrestrial hot springs from the USA, Iceland, China, Italy, and Russia ranged from 40.76 to 50.40%, and their pH values ranged from 1.8 to 7 [31]. The population anammox genome was binned from the metagenome and included 209 contigs, 43.3% GC content with a 3.73 Mbp consensus sequences and an average coverage depth of 27.4 kb [32]. During pile fermentation, a preliminary metagenomic investigation of puer tea produced average read lengths of 456 bp and GC contents of 55% [33]. The sequencing data with a read length of 150 (bp) was obtained from 61,898,482 reads in the Reshi geothermal spring sample and 37,338,796 reads in the Yumthang geothermal spring sample using the Illumina HiSeq. The GC concentrations of Yumthang and Reshi geothermal springs were reported to be 65% and 64%, respectively [34]. For a metagenomic perspective on the geothermal springs microbiomes in the Subantarctic Kerguelen Islands, the GC% varies between 32.15 and 69.21% for only the high- and near-high-quality MAGs, while the GC% ranges between 25.76% and 70.35% for all other MAGs [35]. In our previous study, the high throughput sequencing analysis of Chumathang hot spring, Ladakh, yielded 72,191,530 and 51,097,990 reads, with an average length of 159 bp in soil and water samples with read lengths (in bp) of 135 and 149, respectively. The average GC (guanine-cytosine) content across all contigs were 47% (water), and 58% (soil) [36].

Table 1 Sample quality control and sequence statistics of raw and trimmed data

Attributes	PugW	PugS	PugSo
No. of samples	1	1	1
Raw Data			
Total reads	101,275,022	81,719,124	92,235,980
Total bases	16,102,728,498	12,993,340,716	14,665,520,820
Total GB	16.1027285	12.9933407	14.6655208
Average GC (%)	55	49.5	55
Length (bp)	159	159	159
Total sequences (Millions)	50.6	40.9	46.1
Trimmed Reads			
Total reads	88,749,234	70,790,984	90,785,094
Total bases	13,528,753,711	10,142,633,837	14,365,804,398
Total GB	13.5287537	10.1426338	14.3658044
Average GC (%)	55	49	55
Length (bp)	136	127	149
Total sequences (Millions)	48.1	39.8	44.6
Bacterial phyla (%)	44.8	39.7	41.4
Archaeal phyla (%)	8.6	8.6	8.6
Unclassified (derived from other sequences, %)	27.6	27.6	15.5
Unclassified (derived from bacteria, %)	12	13.8	13.8

The taxonomic classification of high throughput metagenomic sequencing data for the Puga (natural geothermal geyser) sample processed with the KARAKEN2 database showed the domain bacteria as the most abundant domain (PugW: 93.86%, PugS: 98.39%, and PugSo: 98.55%, Fig. 2). MG-RAST analysis of whole genome sequencing showed a total of 44.8%, 39.7% and 41.4% bacterial phyla in the PugW, PugS, and PugSo samples respectively (Table 1) and this analysis also showed a total of 8.6% of archaeal phyla in all the samples. Further, the presence of unclassified (derived from other sequences, PugW: 27.6%, PugS: 27.6%, and PugSo: 15.5%) and unclassified (derived from bacteria, PugW: 12%, PugS: 13.8%, and PugSo: 13.8%) were also observed in the current study. The most prevalent phylum in the PugW, PugS, and PugSo samples was Pseudomonadota (78.37%, 62.42%, and 60.63%, respectively, Supplementary Fig. 1). The prevalent bacterial phylum, the Pseudomonadota, was formerly thought to be solely contain mesophilic species, However, additional species of this

phylum have been identified in different geothermal habitats, such as hot springs [37]. Examples include geothermal regions in Tibet that were studied for diversity of microorganisms [38–39]. The abundance of Pseudomonadota suggests that the microbiota may contribute to photosynthesis, sulfur metabolism, microbial metabolism, methane metabolism, and nitrogen fixation at the site [40]. In the PugW sample, the second most prevalent phylum was Bacteroidota (7.30%) followed by Cyanobacteria (3.81%), Chloroflexota (2.13%), Bacillota (1.78%), etc. The unclassified (derived from eukaryotes, 0.59%), unclassified (derived from Bacteria, 0.22%), and unclassified (derived from Viruses, 0.20%) were also present but in very low quantity.

In the PugS sample, Bacteroidota (18.03%) was the second most populated phyla followed by Bacillota (4.66%), Verrucomicrobiota (3.56%), Actinomycetota (2.66%), Chloroflexota (1.33%), Planctomycetota (1.20%), Cyanobacteria (1.12%), Euryarchaeota (0.94%), Acidobacteriota (0.82%), Chlorobiota (0.47%), unclassified (derived from bacteria)

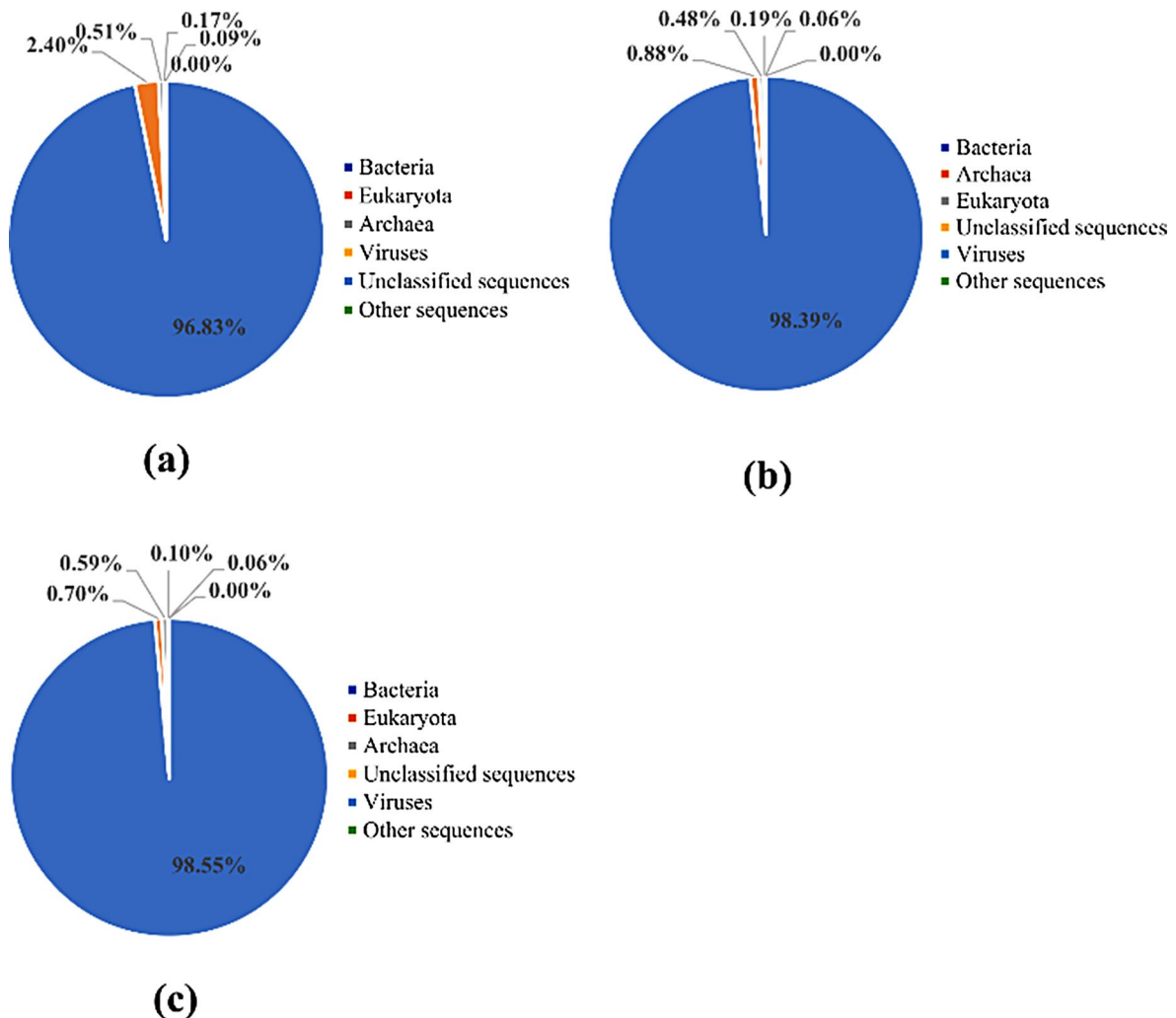


Fig. 2 Pie chart showing the taxonomic distribution of the samples (a) PugW (b) PugS and (c) PugSo at the Domain level

(0.43%), Spirochaetota (0.31%), unclassified (derived from unclassified sequences), Gemmatimonadota (0.23%), Lentisphaerota (0.19%), Deinococcota (0.13%), Fusobacteriota (0.11%), Nitrospirota (0.09%), Thermotogota (0.09%), Streptophyta (0.08%). In the PugSo sample, Bacteroidota (12.22%) dominated after Pseudomonadota, followed by Actinomycetota (8.83%), Bacillota (6.96%), Cyanobacteria (2.47%), Planctomycetota (1.69%), Chloroflexota (1.20%), Deinococcota (1.11%), unclassified (derived from Bacteria) (0.38%), while most archaeal sequences were related to phylum Euryarchaeota (8.6%). Contrary to our data, soil samples from Aguas Calientes geothermal spring in Peru [41] revealed the dominance of five phyla i.e. Proteobacteria (50.09%), Firmicutes (18.73%), Chloroflexi (16.31%), Acidobacteria (3.31%), and Bacteroidetes (2.31%). In Croatia's uncharted geothermal waters [42] and Tengchong, China's terrestrial hot springs [1], the dominant communities of bacteria have been attributed to Proteobacteria and Campylobacteria, whereas the majority of the archaeal sequences have been attributed to Crenarchaeota. In the Philippines' acidic to circumneutral hot springs, Crenarchaeota, Euryarchaeota, and unidentified Archaea made up the bulk of the archaeal phyla [2]. When compared to the Zharkent geothermal spring in Kazakhstan, the phylum Firmicutes was the most abundant, accounting for 89.0% of the total, with Proteobacteria accounting for 5.30% and Actinobacteria accounting for 3.90% [43]. In our previous metagenomics study from the Chumathang hot spring from Himalayan Geothermal Belt, it was found that bacterial and archaeal communities were prevalent in water and soil samples. The study revealed the presence of unclassified (derived from other sequences, soil: 19.05% and water: 17.31%) and unclassified (derived from bacteria, soil: 12.70%, and water: 13.46%) as well as a total of 42.86% (soil) and 50% (water) bacterial phyla and a total of 7.94% (soil), and 9.62% (water) archaeal phyla. Proteobacteria, Firmicutes, Bacteroidetes, Aquificae, and Deinococcus Thermus were found in the water sample. In the soil sample, Bacteroidetes, Proteobacteria, Planctomycetes, Actinobacteria, and Deinococcus Thermus were the predominant microbiological groups. In addition to these bacterial phyla, the samples also included five archaeal phyla: Euryarchaeota, Nanoarchaeota, Korarchaeota, and Thaumarchaeota [36].

At the genera level, *Pseudomonas* dominate in both PugW and PugSo samples but their percentage showed variations (13.82% and 12.87% in PugW and PugSo, respectively, Supplementary Fig. 2). Similarly, *Pseudomonas* was the most prominent genus (0.40%) abundance reported from Zarkent geothermal spring in Kazakhstan [43]. *Pseudomonas* uses hydrocarbons as an energy source for survival and is widely recognized for its capability to digest alkanes, as well as aliphatic hydrocarbon degradation and crude oil

bioremediation [44]. In the PugW sample, the second most dominated genus was *Acinetobacter* (12.87%) followed by *Thiobacillus* (5.30%), *Hyphomonas* (4.27%), *Caulobacter* (4.00%), *Brevundimonas* (3.74%), unclassified (derived from *Flavobacteriales*) (3.07%), *Legionella* (2.42%), *Rhodobacter* (2.21%), *Algoriphagus* (1.81%), *Anaerolinea* (1.78%), *Burkholderia* (0.96%), *Candidatus Solibacter* (0.77%), and *Roseiflexus* (0.74%), etc.

In the PugS sample, the most abundant genera were *Dechloromonas* (15.72%) followed by *Acrobacter* (12.70%), *Geobacter* (5.77%), *Sulfurospirillum* (3.39%), *Bacteroides* (3.17%), *Paludibacter* (2.96%), unclassified (derived from *Verrucomicrobia* subdivision 3, 2.46%), *Pelobacter* (1.41%), *Flavobacterium* (1.41%), *Anaeromyxobacter* (1.25%), *Clostridium* (1.19%), *Anaerolinea* (1.13%), *Opitutus* (0.98%), *Candidatus solibacter* (0.87%), *Candidatus Accumulibacter* (0.81%), *Burkholderia* (0.79%), *Albidiferax* (0.78%), *Pedobacter* (0.75%), *Chthoniobacter* (0.73%), *Prevotella* (0.65%), *Sideroxydans* (0.59%), *Polaromonas* (0.58%), etc. In the PugSo sample, *Psychrobacter* (9.12%) was the second-dominated genus followed by *Allochrochromatium* (2.26%), *Hypomonas* (2.19%), *Algoriphagus* (2.08%), *Bacteroides* (1.86%), *Clostridium* (1.80%), *Planococcus* (1.69%), *Gemmatimonas* (1.59%), *Truepera* (1.17%), *Anaerolinea* (0.92%), *Rhodobacter* (0.88%), *Bacillus* (0.87%), *Streptomyces* (0.86%), *Burkholderia* (0.73%), *Xanthomonas* (0.72%), *Erythrobacter* (0.71%), *Marinobacter* (0.69%), *Brevundimonas* (0.63%). Several economically significant microorganisms, belonging to the genera *Thermus*, *Brevibacillus*, *Bacillus*, and *Pseudomonas*, were found to be prevalent in the geothermal geyser, as reported by previous studies [45, 46].

In the present study, some genera were found abundantly in PugS sample like *Dechloromonas*, *Acrobacter*, *Geobacter*, *Sulfurospirillum*, *Paludibacter*, *Pelobacter*, *Anaeromyxobacter*, *Flavobacterium*, *Opitutus*, *Candidatus Accumulibacter*, *Albidiferax*, *Pedobacter*, *Chthoniobacter*, *Prevotella*, *Sideroxydans*, *Polaromonas*, while in water and soil samples, presence of these genera is significantly low. Similarly, in the PugW sample, the other dominant genera present included *Acinetobacter*, *Thiobacillus*, *Caulobacter*, *Legionella*, and *Roseiflexus* which were present in very low concentrations in PugS and PugSo samples. Further, some genera which included *Psychrobacter*, *Allochrochromatium*, *Planococcus*, *Gemmatimonas*, *Truepera*, *Bacillus*, *Streptomyces*, *Xanthomonas*, *Erythrobacter*, and *Marinobacter* were found abundantly in the soil samples. The most prevalent archaeal genera at the genus level were *Methanosarcina*, *Methanoregula*, and *Methanosaeta*, etc. *Candidatus Aramenus* is an acidophilic and thermophilic genus discovered only in the geothermal site Los Azufres, Michoacán [47]. Furthermore, *Hydrogenobaculum*, a chemolithoautotrophic

At the species level, more than 2000 species were identified, and among the 2000 identified species, only 108 were found most abundant species in all three samples (PugW, PugS, and PugSo, Fig. 3 and Supplementary Fig. 3). The most abundant microbial strains indicated in the inclusive metagenomic study of water, sediments, and soil samples of the Puga geothermal geyser area included *Dechloromonas aromatica* (2.6%), followed by *Acinetobacter baumannii* (1.7%), *Arcobacter butzleri* (1.5%), *Pseudomonas mendocina* (1.4%) and 0.10% each of *Pseudomonas*



fluorescens, *Rhodobacter sphaeroides* and *Pseudomonas aeruginosa* and *Thiobacillus denitrificans*. The abundance of *Dechloromonas aromatica* in individual samples of PugW, PugS, and PugSo were identified as 0.6%, 6.9 and 0.3, respectively. *Acinetobacter baumannii* (PugW: 5.05%, PugS: 0.07%, and PugSo: 0.10%). *Pseudomonas mendocina* (PugW: 2.40%, PugS: 0.15% and PugSo: 1.18%). *Pseudomonas fluorescens* (PugW: 1.51%, PugS: 0.08%, and PugSo: 1.46%). *Rhodobacter sphaeroides* (PugW: 1.86%, PugS: 0.16%, and PugSo: 1.01%). *Pseudomonas aeruginosa* (PugW: 1.50%, PugS: 0.20%, and PugSo: 1.23%), and an abundance of *Thiobacillus denitrificans* (PugW: 2.30%, PugS: 0.22%, and PugSo: 0.22%). Overall, 108 species were plotted for structural analysis of the samples for the comparison study. The most prevalent archaeal species were *Methanosaeta thermophila* (PugW, 0.11%), *Methanoregula boonei* (PugS, 0.14%), and *Methanosarcina berkeri* (PugSo, 0.07%). In a similar study, samples of soil and water from Chumathang Hot Spring in Ladakh revealed a total of 2520 and 1953 species, respectively. The most prevalent species included *Thermus thermophiles*, *Rhodobacter sphaeroids*, *Bacillus cereus*, *Acinetobacter baumannii*, *Bacillus subtilis*, *Aquifex aeolicus*, *Bacillus sp. NRRL B 14,911*, *Hydrogenobacter thermophiles*, and *Algoriphagus sp. PR1*. Comparably, among the archaeal diversity, there are the following: *Aeropyrum prenx*, *Acidilobus saccharovorans*, *Pyrobaculum aerophilum*, *Hyperthermus butylicus*, *Pyrobaculum arsematicum*, *Haloterrigena turkmenica*, *Natrialba magadii*, *Halalkalicoccus jeotgali*, *Haloarcula marismortui*, and *Natronomonas pharaonis*, etc [36]. Since MG-RAST is inefficient for analyzing eukaryotic and viral sequences, the primary focus of the current study was on bacteria and archaea. Thus, we didn't get into them in more detail. It was further observed that only a small percentage of hot springs in the abovementioned geothermal area were subjected to microbiological research. The data is still very scarce, and thus there is a great scope for research in these extreme environments.

Functional profile

MG-RAST is an open-source web application that automatically produces functional assignments and phylogenetic analysis of metagenome. COG, KO, NOG databases were employed to study the functional profile of microbial community. Functionally related genes (orthologs) were analysed by using COG database, while KO analysis demonstrated enzymes onto known metabolic pathways [41]. NOG analysed functional features based on orthology prediction [51]. In the present study, According to the COG analysis, metabolism was the most prevalent category, accounting for 41.82% in PugW, 40.01% in PugS,

and 42.77% in PugSo samples, followed by cellular processes and signalling, which comprised 23%, 27.56%, and 22.43% in PugW, PugS, and PugSo respectively (Fig. 4). In both PugW and PugS samples, the third most abundant functional category was information storage and processing, 17.78% and 16.54% respectively. Similarly, in PugSo, 17.18% of sequences were coded for the information storage and processing category. Additionally, approximately 17.40% of sequences in PugW, 15.89% in PugS, and 17.18% in PugSo sequences were poorly categorized. Metagenome-predicted functional classifications utilizing the NOG database revealed that about 80% of the sequences were poorly characterized in all the samples. In the PugW second dominant functional category is cellular process and signalling, followed by metabolism, and information storage and processing. Metabolism was found second dominant functional category in the PugS sample followed by cellular process and signalling, and information storage and processing. In the PugSo sample, information storage and processing was found to be the second most abundant functional category followed by, metabolism, and cellular processes, and signalling. These findings underscore the diversity and distribution of functional categories within the metagenomes of the studied samples, suggesting variations in microbial community compositions and potential ecological roles across environments. According to the results derived from the KO (Kyoto Encyclopedia of Genes and Genomes) metagenome predicted functions analysis, a predominant proportion of genetic sequences was observed to be implicated in metabolic processes across all samples, followed by genetic information processing, environmental information processing, cellular processes and signalling, human diseases, and organismal systems, in descending order of abundance. To investigate the functionality of the metagenomic data further, genes assigned to the Subsystems functional category was found that the clustering based subsystems was the most abundant category in all the samples (PugW: 14.15%, PugS: 13.79%, and PugSo: 14.17%, Fig. 5). The clustering-based subsystem groups protein families with functional coupling evidence but unknown function, and was also the most abundant subsystem found in the water of the Lobios hot spring in Ourense [52]. The second most dominant category was carbohydrate (PugW: 13.50%, PugS: 13.28% and PugSo: 13.50%), followed by amino acids and derivatives (PugW: 10.34%, PugS: 9.77% and PugSo: 10.60%), protein metabolism (PugW: 7.06%, PugS: 7.39% and PugSo: 6.75%), cofactors, vitamins, prosthetic groups, pigments (PugW: 6.53%, PugS: 6.32% and PugSo: 6.43%), and RNA metabolism (PugW: 5.39%, PugS: 5.71% and PugSo: 5.22%), among the dominant category. Interestingly, genes assigned to photosynthesis and dormancy or sporulation were the lowest categories in all the samples.

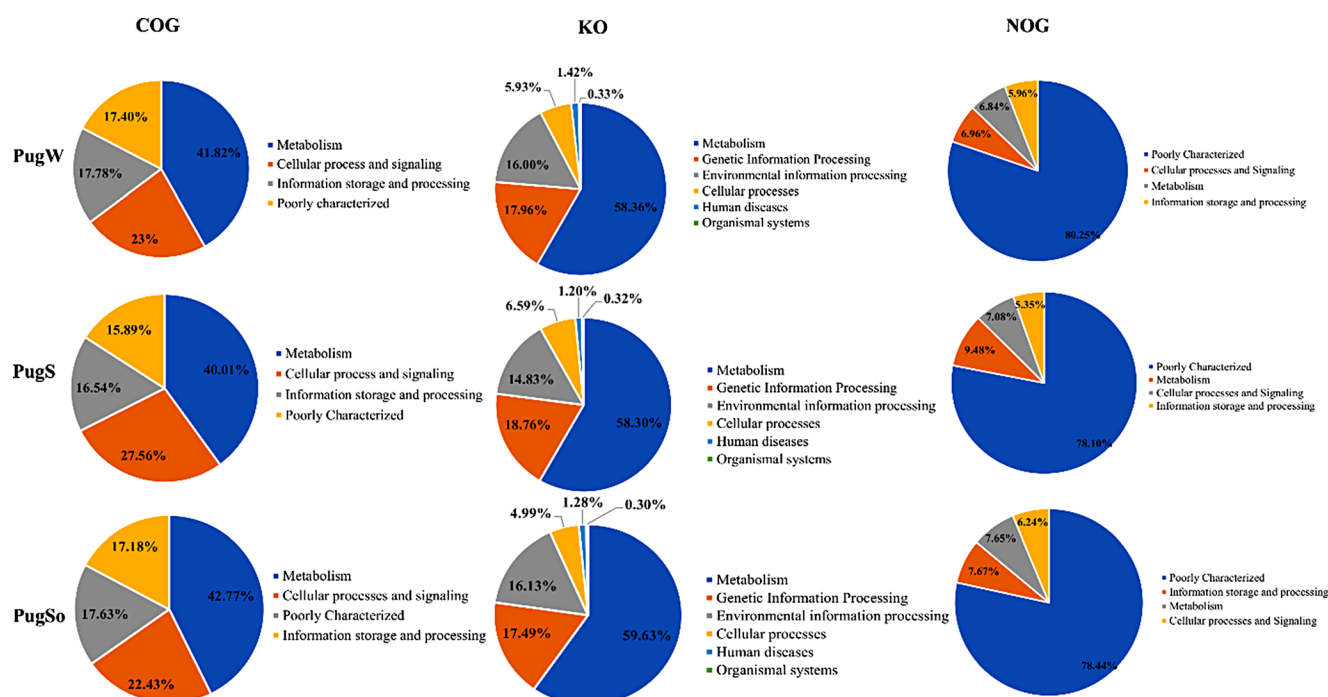


Fig. 4 The pie charts show the distribution of functional groups at the highest level assisted by certain functional hierarchies for COGs, KOs, NOGs. Each slice indicates the percentage of readings for the given source with predicted protein functions annotated to the category

The COG, NOG, and KO database provides an overview of the functional profile of putative genes assigned to general metabolic pathways, whereas the Subsystems functional category, provides comprehensive overview of biological processes. The abundance of genes related to metabolic pathways may be responsible for the survival of the organisms in the extreme environment [53]. In the study, *Pseudomonadota*, *Bacteroidota*, and *Bacillota* taxa were found to be notably abundant. These taxa have been reported to play significant roles in various ecological processes, particularly nutrient cycling and carbon metabolism. Additionally, they are associated with the regulation of stress responses and the facilitation of membrane transport systems within their respective environments [54]. Membrane transporters typically mediate the translocation of various biomolecules, including carbohydrates, lipids, proteins, and inorganic nutrients such as metal ions, across cellular membranes [55]. In contrast to our data, COG analysis of the soil from the Aguas Calientes geothermal spring in Peru represented General and Unknown functional categories, with Amino acid transport and metabolism (8.3%) having the highest representation at the second tier, followed by Energy production and conversion (6.7%) and Transcription (6.6%). KEGG analysis showed that the sequences were assigned to Membrane transport (12.1%), Amino acid metabolism (10.3%), and Carbohydrate metabolism (10.2%) categories [41].

Phylogenetic analysis

Phylogenetic analysis is used to study the evolutionary relationship between biological organisms. It is crucial for understanding biological diversity, genetic classifications, and the developmental processes that take place during evolution. The sequence of a gene is currently used in phylogenetic analysis to identify the evolutionary relationships between species as a result of advancements in genetic sequencing technology. The phylogenetic tree and heatmap were based on the most abundant 50 bacterial species in PugW, PugS, and PugSo samples, and the tree was produced using the NCBI common tree and visualized using the Interactive Tree Of Life based on the top 50 average abundance of the OTUs [26].

To perform the taxonomy abundance from domain to species level MGRAST bar plots were used. Significant variation among OTUs was observed in all samples, according to the heat map (Supplementary Fig. 5) and the phylogenetic trees (Supplementary Fig. 5). The retrieved bacterial populations were from four identified taxa, namely *Pseudomonadota*, *Bacillota*, *Chloroflexota*, and *Cyanobacteria*, were similar to those found in various neutral-pH hot springs at lower altitudes [38]. The phylogenetic analysis indicated that bacteria were more diverse than archaea, and the clone sequences were grouped into 82 bacterial and 41 archaeal OTUs, respectively [38].

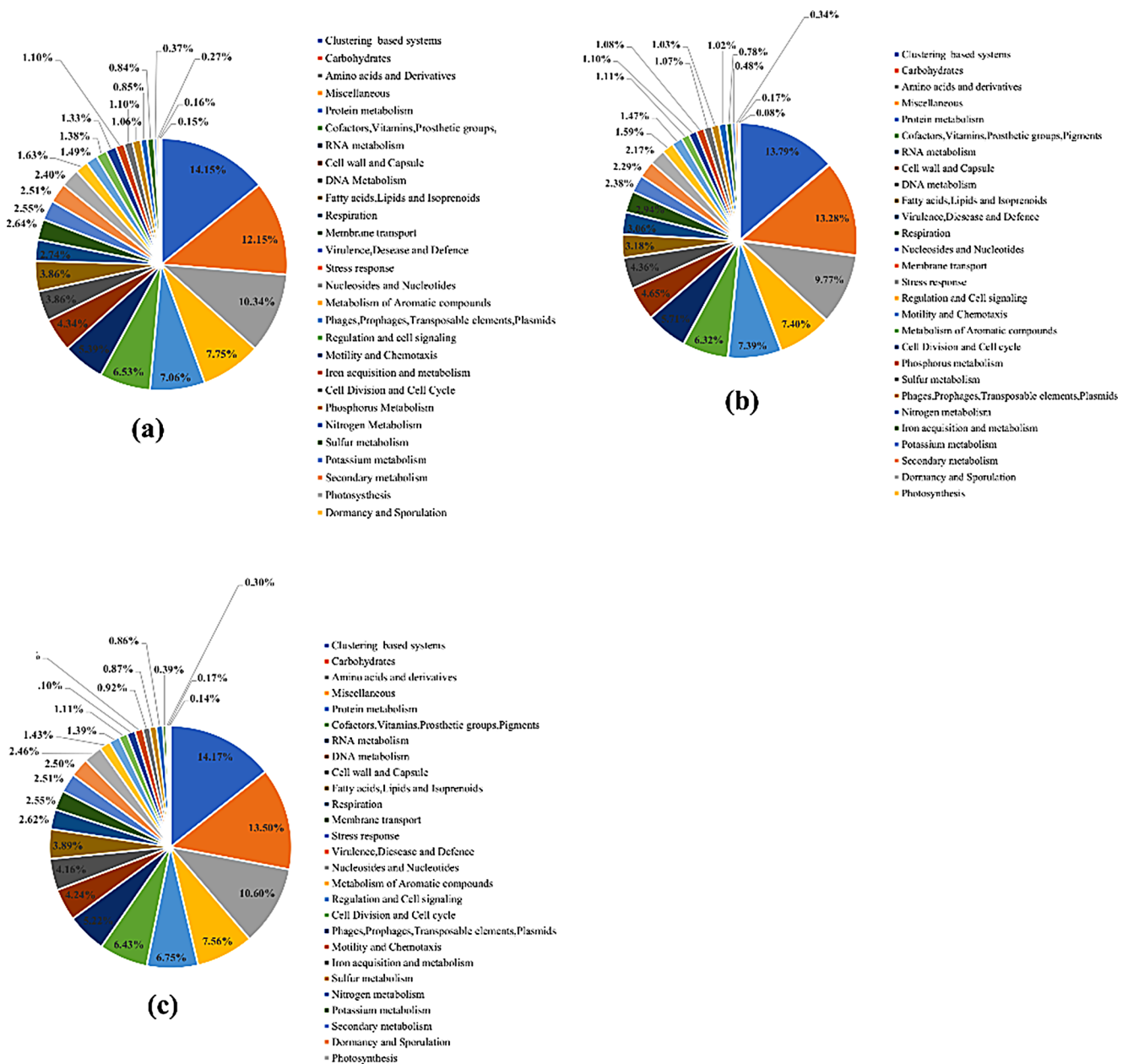


Fig. 5 The pie charts show the distribution of functional groups for the Subsystems for the samples (a) PugW (b) PugS and (c) PugSo. Each slice indicates the percentage of readings for the given source with predicted protein functions annotated to the category

Statistical analysis

A total number of 2362, 2345, and 2468 OTUs were detected in samples PugW, PugS, and PugSo respectively, Shannon–Wiener Diversity (H), Shannon–Wiener Evenness (EH), and Simpson Dominance (D), Indices were used to compare and determine the diversity of microbial communities in the given samples. Supplementary Tables 2 and Fig. 6 illustrate the computed values of the diversity indexes to species evenness and richness. The α -Diversity metrics demonstrated more species diversity in the soil samples as

compared with sediment and water samples. This variation may be due to the abiotic and biotic factors of soil which provide a habitable environment to flourish/support a rich microbial community. The alpha diversity (Shanon) differences between sediment and water samples were connected with physical-chemical variables e.g. water sample equity, DO, sulphate, temperature, and TOC etc [56]. To compare the taxonomic compositions among these microbiomes, we performed Principal Component Analysis (PCA) using the relative abundance data of the genus level. PCA analysis was performed using Biplots and loading plots, The data

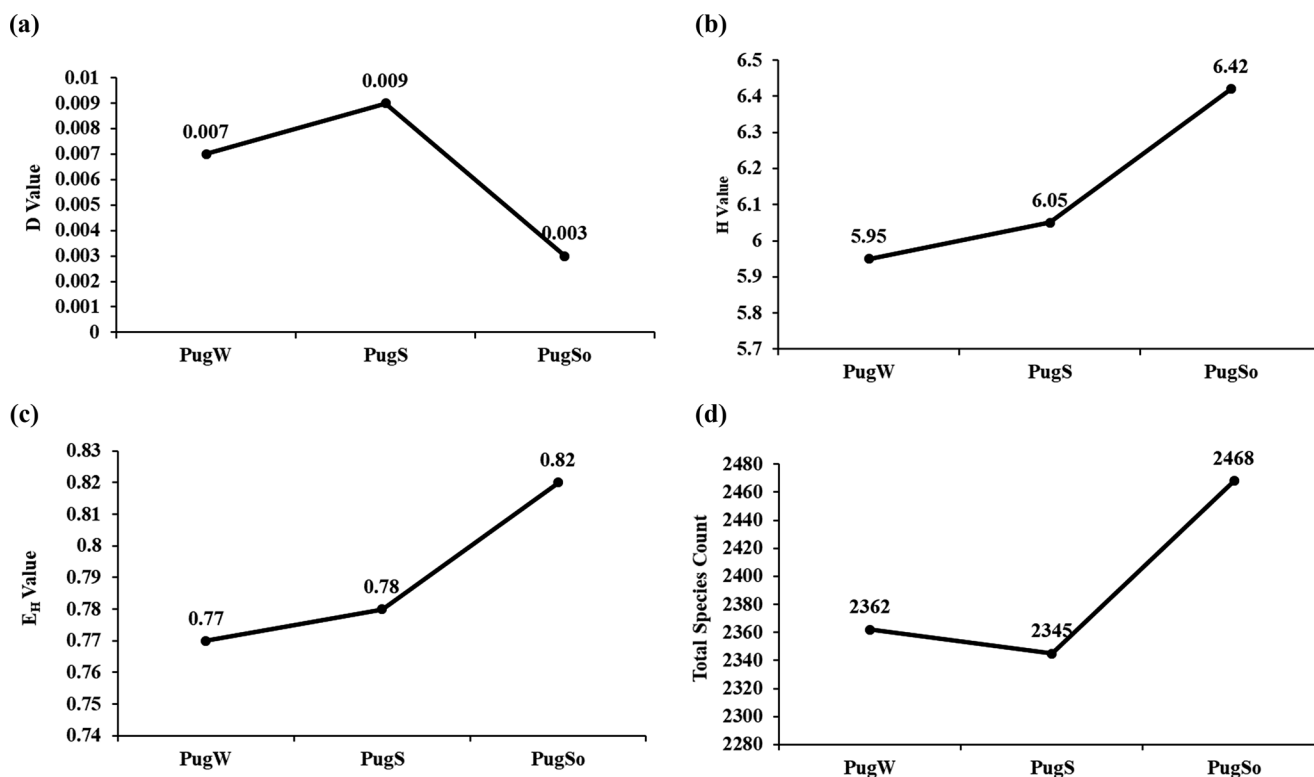


Fig. 6 Simpson Dominance (a), Shannon Diversity (b) and Shannon Equitability (c) Indices, and total species count (d)

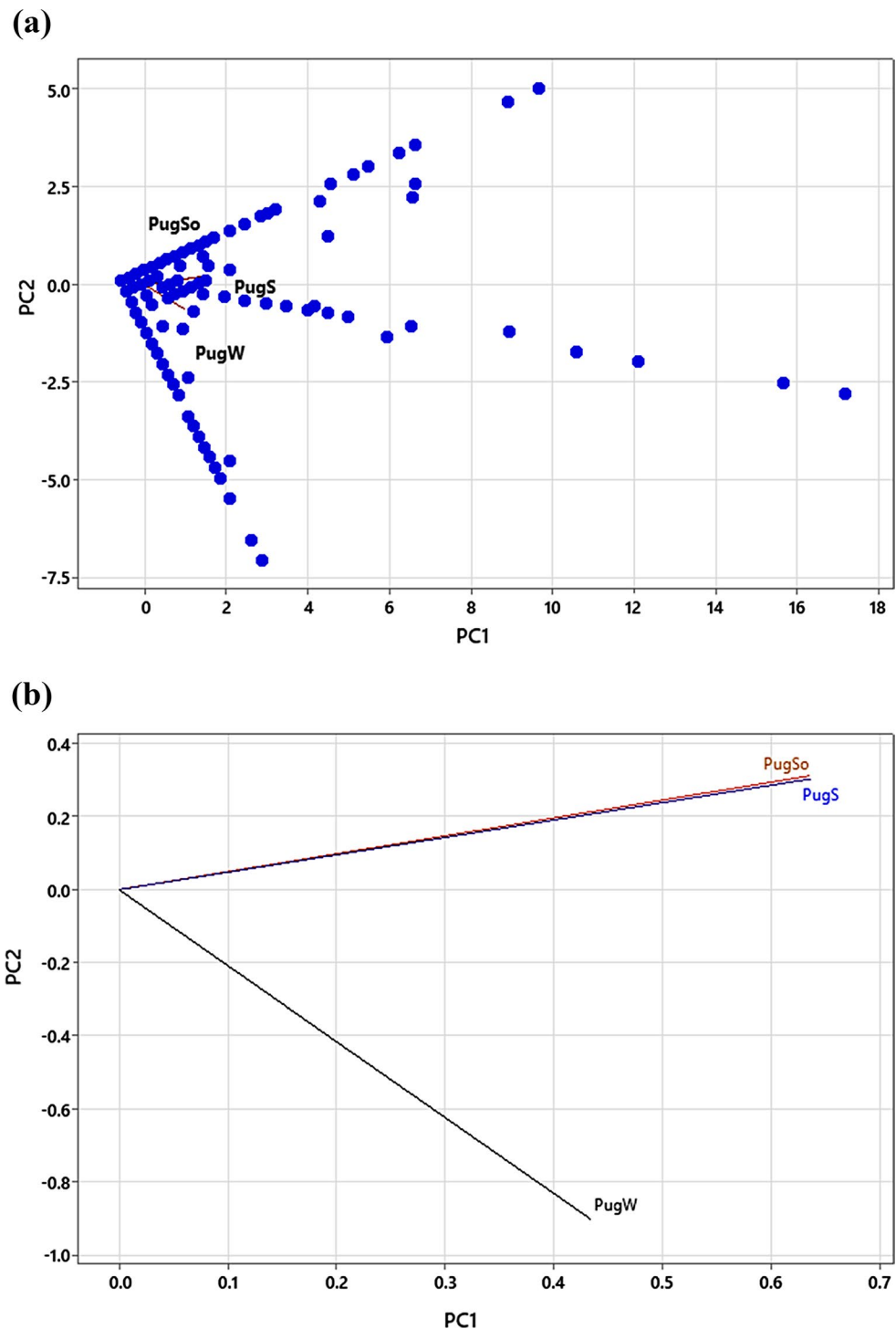
depicted in PCA plotted figures (Fig. 7), PugS and PugSo show a moderate positive correlation and PugW showed a lower positive correlation with the samples.

Conclusion

The present study revealed that the taxonomical diversity and functional analysis of the Puga geothermal geyser and its nearby areas were found to be distinct and may be altered by pH, temperature, and different organic compounds. Pseudomonadota and bacteroidota dominated the bacterial populations in the study location, just like they did in other hot springs reported in the literature. Consequently, the diversity of bacteria was greater than that of archaea. The majority of the archaeal populations in the Puga geothermal geyser and the surrounding area under investigation were made up of the Euryarchaeota clone sequences, that were previously shown to be widespread in other thermal springs throughout the world. This study also revealed the presence of numerous unclassified bacterial taxa, which suggests the possibility of new genera or phylotypes. Various factors, including salinity and pH, were shown to play a substantial role in defining the concentration of the various

taxa in the samples. The functional study of the microbiotas shown that the majority of the genes are linked to metabolic, cellular, signalling, and environmental information processing processes. Furthermore, the investigation revealed the existence of various primary substrate metabolic and biosynthetic routes, including those for clustering based subsystems, carbohydrates, amino acids, miscellences, lipids, proteins, and others, and demonstrating their economic significance. In conclusion, the Puga geothermal geyser and its nearby areas may contain novel biological and genetic materials due to its remote geographic location. Consequently, the current study is being done to examine the thermophilic microbial diversity in the Puga natural geothermal geyser of the Himalayan Geothermal Belt in light of the information mentioned earlier. Previous research in this area had focused on magnetotelluric studies, geology or geochemistry, and geothermal energy. As a result, this work represents the first report on Puga natural geothermal geyser to characterize and identify their microbial diversity at a taxonomic level from phylum up to species level to the best of our knowledge. Future research should make extensive use of metagenomics, metatranscriptomics, and cultural analysis to investigate the physiological traits and adaptive mechanisms of microbes living in this unique ecosystem.

Fig. 7 Principal component analysis (PCA) of the relative abundance (a) biplot and (b) loading plot of each sample at the genus level. The arrows represent the correlation coefficient between the principal component scores



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Author contributions SD provided significant contributions to the manuscript's conceptualization and drafting. SD and SK collected the samples and data to write the manuscript. SD and SK wrote the manuscript, with support from KA and GC. The submitted manuscript was read and approved by all of the authors.

Data availability This research article and its supplementary information file (Supplementary file 1: Doc 1613 kb) contain all of the data generated or analyzed during the course of the investigation.

Declaration

Competing interest The authors declare that they do not have any known competing financial interests or personal ties that could appear to have influenced the work disclosed in this study.

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